RED FM

Weekly checks

Border Holdings, Avonmouth

Date: 2026-11-11

Ref:	WK-MWE1H3EL
Date:	2026-11-11
Engineer:	Stephen Norley
Plant:	A & B, C & D, E
Week:	46
Ambient °C:	8.5°c
Faults recorded:	0

Faults recorded on this visit

None - no line item was flagged.

A & B - Chamber temperature

Item	A1	A2	A3	A4	No1 B2	No2 B2	No3 B1	No4 B1
Cold store set point (controller)	-20	-20	-20	-20	-18.9	-18.9	-18.9	-18.8
Cold store temp air-on	-20.6	-19.8	-20.8	-20.2	-17.7	-17.7	-17.8	-18.7
Coolers air off	-22.8	-21.6	-22.8	-22.1	-19.6	-17.9	-19.7	-18.4
Cooling on/off	On	On	On	On	On	On	On	On

A & B - Coolers

Item	A1	A2	A3	A4	No1 B2	No2 B2	No3 B1	No4 B1
Cooler fan correct operation	OK	OK	OK	OK	OK	OK	OK	OK
Cooler correct operation	OK	OK	OK	OK	OK	OK	OK	OK
Cooler correct defrosting cycle	OK	OK	OK	OK	OK	OK	OK	OK
Cooler ice build-up & debris	OK	OK	OK	OK	OK	OK	OK	OK
Defrost drain line for leakage	OK	OK	OK	OK	OK	OK	OK	OK
Operating suction superheat #	7.3	8.4	8.2	8.5	7.2	6.4	6.5	8
% opening of expansion valve #	61	62	46	44	64	25	100	5
Cooler air temp diff #	2.2	1.8	2.0	1.9	1.9	0.2	1.9	-0.3

A & B - Compressor racks & condenser

Item	A	B
Compr abnormal noise / vibration	OK	OK
Oil cooler fin face free of debris	OK	OK
Oil cooler fans correct operation	OK	OK
Plant-room ventilation fan (on/off)	Not wired in	Not wired in
Plant-room temperature	18.8	18.8
Wet-matt cooling on/off (hours)	Off	Off
System delivery pressure	16.6bar	12.7bar
Condenser fin face free of debris	OK	OK
Condenser fans correct operation	1x condenser fan faulty no3	Operating correctly
Liquid line sight glass	Full	Full
Receiver level	Above bottom sight glass	Above bottom sight glass
Oil separator oil level	Full	Full
No1 compressor running & % load	Off 0%	On 50%
No2 compressor running & % load	On 100%	Off

Item	A	B
No3 compressor running & % load	On 50%	N/A
No1 compr delivery & econ temp	Comp off	Delivery 65°c econ off
No2 compr delivery & econ temp	Delivery 72°c econ 8°c	Comp off
No3 compr delivery & econ temp	Delivery 68°c to con off	N/A
Refrigerant leak detection system	OK	OK

A & B - Controls

Item	A	B
Suction pressure	O.8bar	1bar
Suction temperature	-19.2	-12
Suction SH	8	12
Delivery pressure	16.6bar	15.5
Common delivery temp	74°c	67.3
Oil temperature	42.3	46.4
HP liquid temperature	33°c	30
HP liquid sub-cooling	7°	8°c
Econ liquid temperature	18	13
Econ sub-cooling	22	25

A & B - Main incoming meter

Item	kWhr
Electric kWhr meter reading	725514
Previous meter reading	716614
Total kWhr	8900

A & B - A - pack meter

Item	kWhr
Electric kWhr meter reading	4705413
Previous meter reading	4686195
Pack total kWhr	19218
A chamber total energy	36983

A & B - B - pack meter

Item	kWhr
Electric kWhr meter reading	2919743
Previous meter reading	2904816
Pack total kWhr	14927
B chamber total energy	22670

A & B - A & B cooler energy meter

Item	kWhr
Electric kWhr meter reading	2687417
Previous meter reading	2671909
Total kWhr	15508

A & B - B cooler energy meter

Item	kWhr
Electric kWhr meter reading	732660
Previous meter reading	724917
Total kWhr	7743

A & B - A cooler energy (derived)

Item	kWhr
Electric kWhr meter reading	1954757
Previous meter reading	1946992
Total kWhr	7765

C & D - Chamber temperature

Item	C1	C2	C3	C4	D1	D2
Cold store set point (controller)	-18.8	-18.8	-18.8	-18.8	-18.6	-18.6
Cold store temp air-on	-18.4	-18.8	-19.5	-19.2	-18.9	-18.9
Coolers air off	-22.4	-19.8	-19.5	-19.5	-21.1	-18.9
Cooling on/off	On	On	Off	Off	On	Off

C & D - Condensing units

Item	C1	C2	C3	C4	D1	D2	D3	D4
Compr abnormal noise / vibration	OK	OK	OK	OK	OK	OK	OK	OK
Compressor abnormal frosting	OK	OK	OK	OK	OK	OK	OK	OK
Compr oil level	Full	Full	Full	Full	Full	Full	Full	Full
Compr delivery temperature	92	65	78	83	78	81	74	69
Oil recovery / separator level	Full	Full	Full	Full	Full	Full	Full	Full
Condenser fin face free of debris	OK	OK	OK	OK	OK	OK	OK	OK
Condenser fans abnormal noise	OK	OK	OK	OK	OK	OK	OK	OK
Liquid line sight glass	OK	OK	OK	OK	OK	OK	OK	OK
Receiver level	Below bottom sight glass	Above bottom sight glass	Above sight glass	Above bottom sight glass	Below bottom sight glass	Below bottom sight glass	Below bottom sight glass	Below bottom sight glass

C & D - Coolers

Item	C1	C2	C3	C4	D1	D2
Cooler temperature diff	4	1	0	0.3	2.2	0
Cooler correct operation	OK	OK	OK	OK	OK	OK
Cooler correct defrosting cycle	OK	OK	OK	OK	OK	OK
Cooler ice build-up & debris	OK	OK	OK	OK	OK	OK
Defrost drain line for leakage	OK	OK	OK	OK	OK	OK

C & D - Energy readings

Item	C2-4 refrig	C1-3 refrig	D 1&2	D 3&4 compr
Electric kWhr meter reading	3050333	1144487	2682461	24738
Previous kWhr meter reading	3040771	1140720	2670599	24653
Total kWhr	9652	3767	11862	85
Chamber total energy	13419		11947	

E - Chamber temperature

Item	E1	E2
Cold store set point (controller)	-18.6	-18.6
Cold store temp air-on	-19	-19.1
Coolers air off	-20.4	-19
Cooling on/off	On	Off

E - E chamber front-centre temp

Item	°C
Front-centre temperature	-18.9

E - Condensing units

Item	E1 Compr1	E1 Compr2	E2 Compr1	E2 Compr2
Compr abnormal noise / vibration	OK	OK	OK	OK
Compressor abnormal frosting	OK	OK	OK	OK
Compr oil level	Full	Full	Full	Full
Compr delivery temperature	64	62	56	54
Oil recovery / separator level	Above top sight glass	Above top sight glass	Above top sight glass	Above top sight glass
Condenser fin face free of debris	OK	OK	OK	OK
Condenser fans abnormal noise	OK	OK	OK	OK
Liquid line sight glass	OK	OK	OK	OK
Receiver level	Above bottom sight glass	Above bottom sight glass	Below bottom sight glass	Below bottom sight glass

E - Coolers

Item	E1	E2
Cooler temperature diff	1.4	-0.1
Cooler correct operation	OK	OK
Cooler correct defrosting cycle	OK	OK
Cooler ice build-up & debris	OK	OK
Defrost drain line for leakage	OK	OK

E - System readings

Item	E1	E2
Suction pressure	0.8	0.8
Suction temperature	-22	-21
Suction superheat	7	6
Delivery pressure	12.0	10.5
Liquid refrig temperature	26	20
Liquid sub	4	6

E - Energy readings

Item	E1	E2
Electric kWhr meter reading	14588	12833
Previous kWhr meter reading	14487	12786
Total kWhr	101	47
Chamber total energy	148	

RDM alarms at time of visit

Item	Asset / chamber	Active since	Outcome	Action taken / reason
S02 stage fault 1	Pack A	2026-11-11	Ongoing	Condenser is being replaced